

GUIDELINES ON DESIGN OF ROAD TRAFFIC SIGNALS

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I. GENERAL

1. INTRODUCTION

The purpose of this publication is to give general guidelines for the design of optimal road traffic signals for the control of traffic.

2. SCOPE

The scope is limited to pre-timed (fixed-time) traffic signals.

3. TERMS AND DEFINITIONS RELATED TO ROAD TRAFFIC SIGNALS

Traffic Signal Control - A power-operated traffic control device by which traffic is alternately directed to stop and proceed in a specified direction at intersections.

Approach - That section of road, consisting of one or more lanes, used by vehicles approaching an intersection.

Signal indication/display - An optical system (circular, arrow, or symbolic) capable of being illuminated at a given time with red, green, or yellow colour.

Signal Face - Part of a traffic signal provided for controlling one or more traffic movements on a single approach.

Signal Head - An assembly containing one or more signal faces placed together facing traffic from one direction.

Flashing Red - When the red lens of a signal face is illuminated by rapid intermittent flashes, drivers of vehicles shall stop before entering the intersection.

Flashing Yellow - A traffic signal indication for which the yellow lens is illuminated with rapid intermittent flashes to signify that drivers of vehicles may proceed through the intersection with caution.

Controllers - A complete electrical mechanism for controlling the operation of a traffic signal.

Manual Controller - A traffic signal controller that is manually operated.

Automatic Controller - A traffic signal controller that is operated automatically.

Pre-timed (Fixed Time) Signal - A type of traffic signal that directs traffic to stop and permits it to proceed in accordance with predetermined time durations.

Cycle - One complete sequence of signal indications.

Cycle time or Cycle Length - Total time to complete one sequence of signalisation for all movements at an intersection.

Interval - Part of the cycle length during which all indications remain constant.

Green interval - The duration, in seconds, of the green indication for a given movement at a signalised intersection.

Red interval - The period in seconds, in the signal cycle, during which the signal is red for a given lane group.

Yellow or change interval - An indication warning the user that one indication has ended, and the next indication will begin.

All-Red or Clearance Interval - The period in which all the signal faces controlling vehicular traffic at an intersection give a red indication.

Inter-green Period (Yellow plus all-red or Change and clearance interval) - The time from the end of the green period of the phase losing right-of-way to the beginning of the green period of the phase gaining right-of-way.

Interval Sequence - A predetermined order of appearance of traffic control signal indications within a cycle.

Signal Phase - Part of the cycle allocated to one or more movements having the right-of-way simultaneously. In other words, it is the green change and clearance intervals in a cycle that are assigned to an independent traffic movement or a combination of traffic movements, e.g., through movement and right-turn movement of an approach moving together.

Vehicular Phase - A traffic phase allocated to vehicular traffic.

Pedestrian Phase - Traffic phase allocated exclusively to pedestrian movement.

Conflicting Movements - Movements at a junction which cannot proceed safely at the same time, e.g., North-South through and East-west through movements.

Conflicting Phases - Phases that control conflicting movements.

Phase diagram – A diagram showing all movements made in a green phase within a single block of diagram

Ring diagram - Diagram showing which movement is controlled by which ring of the controller.

Ring and Barrier - The ring separates phases that may operate one after another. A change interval and clearance time are used to separate that movement in time. A barrier is used to separate the east-west movements from the north-south movements to avoid operating conflicting movements at the same time. Figure 1 shows typical phase and ring diagrams with ring and barriers marked.

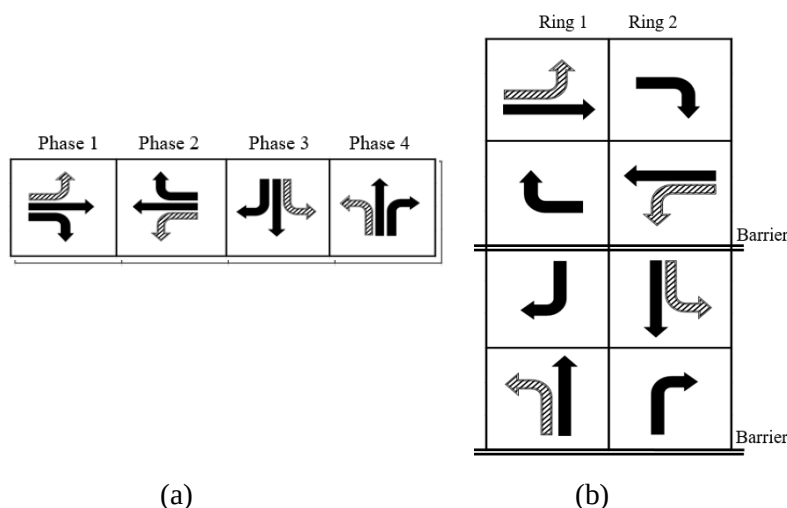


Figure 1. (a) Phase diagram, (b) Ring and barrier diagram

Protected Movement - A signalled movement where conflicting movements are held at red, especially for right turners. Protected movements are shown as solid arrows in phase and ring diagrams, as shown in Figure 1.

Permitted Movement - A signalled movement where conflicting movements are allowed to go together, mostly right turners allowed to move through the opposite through when they can find safe

gaps. Permitted movements are shown as dotted arrows in the phase and ring diagrams, as shown in Figure 2.

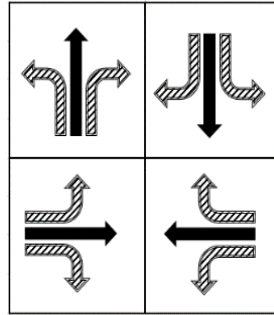
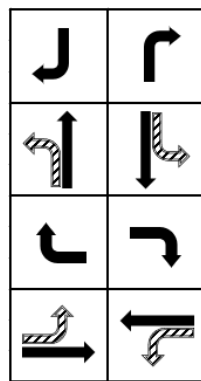
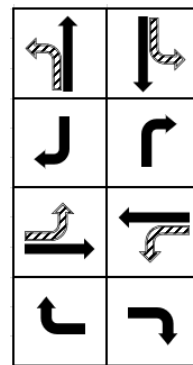


Figure 2. Sample ring diagram with permitted movements

Concurrent Phasing – A phasing arrangement where the right-turn movements of opposing directions are together in one phase, while the opposing through movements are kept in another phase. This can be of two types, namely leading or lagging, based on whether the turn phase is before or after the through phase. Figure 3 shows a ring diagram for these phasing arrangements.



(a) leading right



(b) lagging right

Figure 3 Concurrent phasing

Lead-lag phasing - In this phasing, one right turn leads the through movement while the other lags the through movement. Figure 4a shows a ring diagram of a lead-lag phasing.

Split phasing - Split phasing is provided in locations where the simultaneous turn movements conflict due to space constraints and hence cannot be provided together. Thus, in this arrangement, all movements of each approach move together, followed by all movements of another approach. Figure 4b shows a typical split phasing.

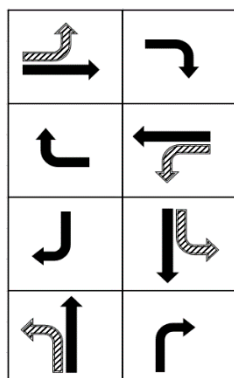


Figure 4(a) Lead-lag right-turn phase

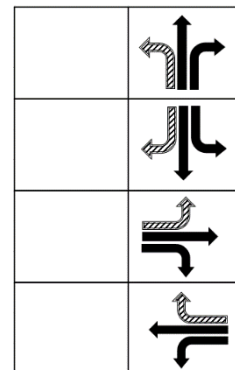


Figure 4(b) Split phasing

Late Release - During the same phase, when one particular movement/ set of movements is allowed to take place a little later than the actual start of the phase.

Early Cut-off - During the same phase when particular movement/movements are terminated before the actual end of the phase.

Overlapping and Non-overlapping phasing - When the traffic volumes of the movements operating within the same phase are not comparable in magnitude, an overlap phase, where the movements may not start and end together, is adopted. Figure 5 shows a typical ring diagram for an overlapping phase. Note that otherwise, the movements start and end together, called non-overlap, as shown in Figures 3 and 4a.

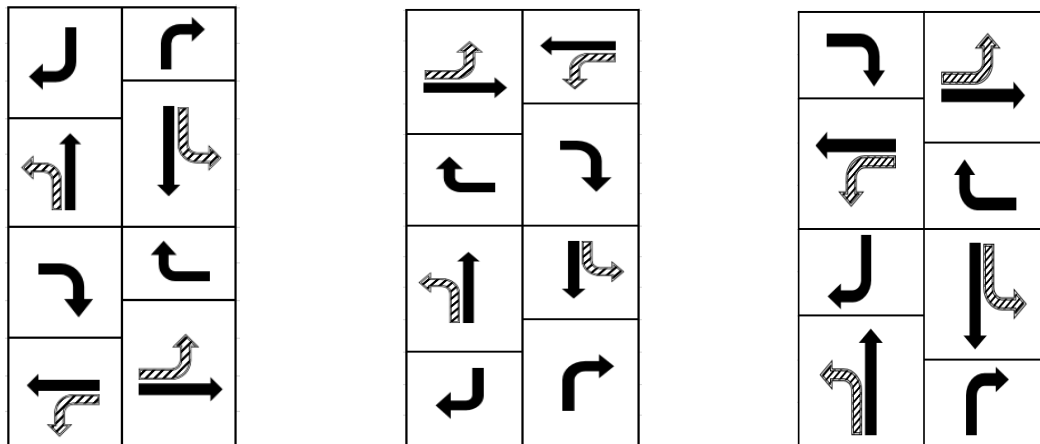


Figure 5 Typical overlapping phase diagrams

More advanced phasing options, such as compound phasing and associated challenges, such as the yellow trap, are not considered in this manual.

Manual Operation - The operation of an automatic controller mechanism by means of a hand-operated switch.

Emergency Features - A relay or other device installed in a controller that, when activated from a remote point, discontinues the regular signal operation and displays special signal indications for the movement of an emergency vehicle, etc.

Isolated Control - Control of a signal installation where the timings are not related or linked to adjacent intersection signal installations.

Vehicle Actuation (VA) - A method of control where the green times are extended or truncated based on the real-time flow of traffic on the approaches. Detectors are used to monitor the traffic flows.

Signal Coordination - An arrangement that relates the timings of a signal to the timings of a neighbouring signal.

Offset - The time difference between a specific point in the cycle at a junction and a reference point, such as the start/end of a phase at an adjacent junction.

Green Wave - A control strategy for a linear system of traffic signals that synchronises the start of green with the arrival of the platoon from the upstream junction.

Platoon - A group of vehicles that move together as they get discharged from a signal when the green starts after the red.

Area Traffic Control System - This involves the coordination of traffic signals over a complete network of signals covering an area that may be considered homogeneous from the point of view of traffic operation. Such a system has to be computer-aided as the problem is highly complex.

Lost time - The portion of time at the beginning of each green period and during the change and clearance period that is not effectively utilised by vehicular flow.

Effective green time - The time during which a given traffic movement or set of movements is utilising the green effectively; it is equal to the green plus yellow time minus the lost time.

Effective red time - The time during which a given traffic movement or set of movements is effectively stopped; it is equal to the cycle length minus the effective green time.

Timing diagram - Diagram showing different intervals involved within a cycle. Figure 6 shows a sample timing diagram for a three-phase signal.



Figure 6 Sample signal timing diagram

Green Split - The division of available green time between phases within a signal cycle.

Discharge headway - Headway between successive vehicles negotiating an intersection during the green time of signal operation.

Saturation headway - Minimum time headway at which vehicles depart one after the other under the queue discharge condition.

Saturation Flow Rate - Theoretical maximum hourly rate at which vehicles can traverse an intersection, assuming continuous green indication at all times and no loss time, in vehicles per hour.

Capacity - The maximum rate at which vehicles can pass through an intersection under prevailing conditions.

Through Car Units (TCU) - Equivalency factor to convert vehicle turning movements to through movement.

Delay - The additional travel time experienced by a driver, passenger, or pedestrian due to the presence of a signal.

Control Delay - The amount of additional travel time experienced by a user attributable to a control device.

Stopped Delay - The amount of time spent by a vehicle at an intersection in a completely stopped condition.

Flow Ratio - It is the ratio of volume to saturation flow used to normalise for comparison across various roads.

Degree of Saturation - Ratio of volume to capacity that is a measure of how close the demand is to the intersection's capacity.

Critical lane - These are lanes with max. (v/s) ratio for each phase.

Lane groups - One or more lanes of an intersection approach operating in stable equilibrium. Some general guidelines for lane grouping are: if turning vehicles are sharing lanes, all lanes in the approach can be considered as a single-lane group, approaches with exclusive right or left turn lanes, or both, can be considered as a lane group and all other lanes can be included as a single-lane group, if an approach with more than one lane includes a lane that may be used by both right-turn and through vehicles, a decision should be made as to whether equilibrium exists or is dominated by right-turn movement.

Virtual lanes - Imaginary lane divisions used to analyse traffic behaviour in non-lane-based conditions based on the number of parallel vehicle movements.

II SIGNAL TIMING DESIGN

A pre-timed/fixed time signal design involves the following steps:

- Identifying the most appropriate phase plan,
- Calculation of the yellow and all-red interval,
- Determination of optimal cycle length,
- Proportioning of green time for each movement,
- Minimum green based on pedestrian requirement

The detailed methodology for these steps is discussed in the following sections.

1. PHASE PLAN

A signal phasing plan refers to the sequence of green indications assigned to different traffic movements at an intersection. The first decision to be taken under this step is whether the right turns need protection or not. If there is a need for a protected right turn, the next decision is between split, lead-lag, and concurrent phasing. This will be followed by the decision to have an overlapping phase or a non-overlapping phase. All these phase design decisions are guided by several factors, including traffic volumes, vehicle composition, lane configuration, turning radii, and intersection geometry. The criteria for deciding each of these signal phasing types are discussed below.

1.1 Protected vs. Permitted

There are two criteria for determining when a protected phase should be provided instead of a permitted phase for right-turning movements at intersections. The first criterion considers the right-turn volume, and the second criterion is based on the cross product of the right-turn and opposing through volumes. The factors influencing the thresholds for these criteria include the composition and number of lanes in the opposite direction. If either of these criteria is met, a protected phasing should be used. Based on simulation studies conducted at IIT Madras, the suggested thresholds for various traffic scenarios are given in Table 1.

Table 1. Thresholds for Protected Phase for RT movements^{1,2}

Bike Proportion ³	Only One Opposing Through Lanes (3.5m)		Two Opposing Through Lanes (7m) ⁵	
	Criteria 1	Criteria 2	Criteria 1	Criteria 2
	Min. RT Volume (Veh/h) ⁴	(RT Vol) x (TH Vol.)	Min. RT Volume (Veh/h)	(RT Vol) x (TH Vol.)
≤ 20%	475	4,00,000	275	2,80,000
20% - 30%	500	4,75,000	325	2,90,000
30% - 40%	550	5,05,000	325	3,15,000
40% - 60%	575	5,75,000	325	3,25,000
60% - 85%	600	6,45,000	375	4,00,000
≥ 85%	650	8,00,000	400	5,50,000

¹In a permitted phase, turning movements from opposite directions are expected to move without conflicts.

²Use the protected phase for the right-turning vehicles if there are more than two lanes in the opposite direction.

³Assuming up to 5% heavy vehicles and up to 25% three wheelers. If the field-observed vehicular composition is not listed in the table, use the closest composition from the table.

⁴For volumes in between, use table values with appropriate interpolation. See the sample application given in Appendix II for further clarification.

⁵Use total flow on both lanes.

1.2 Split Vs. Concurrent and Lead-lag phasing

Based on the analysis of protected versus permitted movements, if the results indicate that the right-turning movements require protection, the next step in the phase plan design process is to determine whether the protected phase is to be split, lead-lag, or concurrent. If there is a space constraint making concurrent right turnings difficult, split phasing should be provided, where all movements of each approach move together. Else, concurrent or lead-lag phasing can be implemented. While lead-lag phasing has the right-turn movement along with its corresponding through movement, the concurrent phasing has the right-turn movements of opposing approaches move simultaneously. Thus, the choice between split vs. concurrent and lead-lag phasing is primarily governed by the geometry of the intersection.

The turning radius of the design vehicle is 12.2 m, and the minimum throat width for the traffic lane under Indian conditions is taken as 1m. Based on these, the diagonal distance between the centres of two turning radii should be at least 25.5 m ($12.2+1+12.2$) as shown in Figure 7. If the available distance in the field (the diagonal distance calculated from the stop line to stop line distances of opposing approaches) is less than the above threshold of 25.5 m, split phasing should be adopted instead of the concurrent and lead-lag phasing.

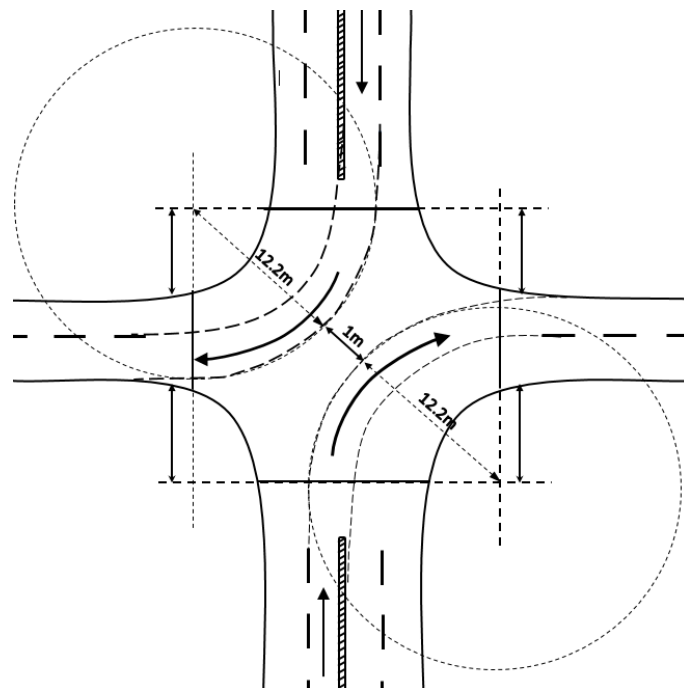


Figure 7 Geometrical specifications for concurrent phasing

When a dedicated right-turn lane is provided, either lead-lag or concurrent phasing may be used. In the absence of dedicated right-turn lanes, lead-lag phasing is recommended. The choice between concurrent and lead-lag phasing depends on the relative traffic volumes. Within lead-lag operation, a lead phase is preferred when right-turn demand is high, while a lag phase is preferable when through traffic dominates.

1.3 Overlap vs. Non-Overlap Phasing

Based on the volume of movements within a phase, the phase sequence can be implemented in two ways: non-overlap and overlap. In non-overlap phasing, all movements within a phase start and end together (see Figure 3 or 4a). The non-overlap approach, although simple and widely adopted, lacks the flexibility to respond effectively to varying traffic demands.

In contrast, overlap phasing allows movements that have lower demand to end early so that their conflicting movement from the opposite approach can start early (see Figure 5). Overlapping green times are assigned to movements that do not conflict with each other, enabling more efficient use of the available green time. The decision to use an overlapping phase is a design choice guided by the engineering judgement and a detailed analysis of traffic flow patterns and safety considerations. A suitable general guideline is to go for the overlap phase when green time requirements for overlapped movements differ by more than 5 s or the movement-wise traffic volume in two approaches differ by more than 60 pcu/hr/virtual lane.

More advanced phasing options, such as compound phasing, and associated challenges, such as the yellow trap, are not considered in this manual.

A flowchart illustrating the complete procedure for phase plan selection is presented in Figure 8.

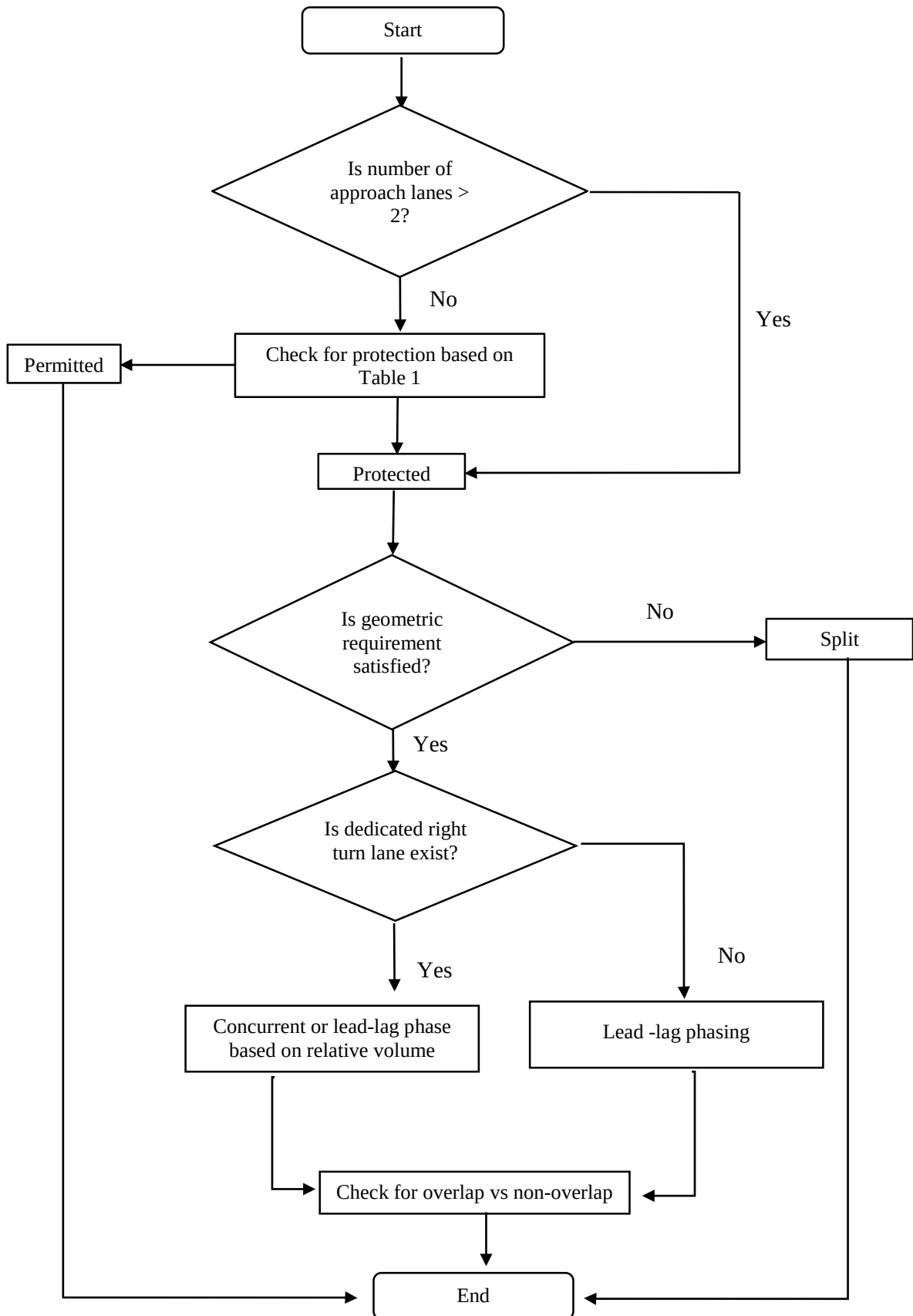


Figure 8. Flowchart of the phase plan selection process

A sample calculation illustrating the phase planning is given in Appendix II.

2. DETERMINATION OF YELLOW (AMBER) AND ALL RED TIME

The change and clearance interval, or the amber plus all-red time, is provided immediately after the green phase for a movement. The minimum amber plus all red time (A) required to avoid dilemma zones at the intersection is given by

$$A = t + \frac{v_0}{2a} + \frac{W+L_0}{v_0} \quad (1)$$

where t – reaction time of drivers at a signalised intersection, generally assumed as 1 s.

v_0 – Average approach speed of vehicles (m/s), take 10 m/s if no information is available

a – Acceleration of vehicles (m/s²); take 3 m/s² if no information is available

a_g - Acceleration due to gravity (9.8 m/sec²)

W - Intersection width (m)

L_0 - Length of standard vehicle (m), take 5 m if no data is available

III. CYCLE TIME CALCULATION

Based on the research conducted at IIT Madras, the following equation is recommended for the calculation of cycle length under mixed traffic conditions. The equation for the Cycle time is derived to minimise delay using queueing theory. As detailed in Appendix I, the queue at the intersection, based on departure distribution, can be M/D/n or M/M/n. The departure distribution can be assessed by counting the number of vehicles discharging during queue clearance, followed by a Chi-square test. In case such data is not available, the following guidelines can be used:

- (a) When motorised two-wheelers dominate the traffic stream (more than 40 percent), the departure distribution is predominantly random, following M/M/n queue discipline. The cycle length is given by

$$C_0 = \frac{1.9 L + 3.2}{1 - Y} \quad (2)$$

- (b) When cars constitute the majority of the traffic composition (more than 40 percent), the departure distribution tends to be deterministic following M/D/n queue discipline.

$$C_0 = \frac{1.6 L + 2.4}{1 - Y} \quad (3)$$

where C_0 - Optimum cycle length in seconds

L - Total lost time per cycle

Y – Sum of critical flow ratio (flow/saturation flow) in each phase

Table 2. The lost time values for heterogeneous and lane-less traffic*

Factors	Characteristics	Lost time per phase in seconds**
Operating conditions	Peak period	2
	Off-peak period	2.2
	Protected turning phase	2.16
Weather condition	Clear weather	2
	Rain, fog, snow	2.5

*Adopted from Agent and Crabtree, 1983; Li and Prevedouros 2002; Anusha, Verma and Kavitha, 2013; Sun et al. 2013; Adams, Zambang and Opoku-Boahen, 2015; Zhang et al., 2023.

**If multiple characteristics are met, choose the highest value.

To calculate the flow ratio, it is essential to estimate the saturation flow.

Estimation of saturation flow

The saturation flow can be estimated by counting the number of vehicles crossing the stop line during the green phase until the queue clears, and converting this to an hourly rate.

To incorporate heterogeneity in traffic, the Passenger Car Unit (PCU) values specified in IRC: 106-1990 can be employed. The turning volumes can be converted to through equivalent volumes using the TCE (Through Car Equivalent). TCE for Indian conditions was derived for both right and left turns and is given in Tables 3 and 4, respectively. In case the measurement of saturation flow is cumbersome, it can be estimated using the empirical relation of 900 TCU/h/unit width of virtual lane, which is derived based on field observation and applicable to both narrow and wide approaches.

Table 3 TCU for Permitted Right Turn

Opposing flow (veh/hr)	No. of opposing lanes	
	1	2
800	3.0	2.9
1000	3.6	3.7
1200	4.1	4.3
1400	4.6	4.9
1600	4.9	5.7
1800	5.7	6.4
2000	6	6.9

Table 4 TCU for left turn against pedestrian volume

Number of conflicting pedestrians(ped/hr)	TCE
None (0)	1.5
Low (50)	1.64
Moderate (200)	1.79
High (400)	1.97
Extreme (800)	2.34

TCE for protected right turning vehicles = 1.2

Calculation of number of virtual lanes

The virtual lane concept is utilised to account for the lane-less movement of vehicles. In HLL traffic conditions, parallel vehicle movements result in the number of servers differing from the number of lanes. The number of virtual lanes is determined by dividing the approach width by the sum of the weighted average width (weighted based on the number of vehicles passing the stop line simultaneously) and the side clearance on both sides of the vehicles (0.53m on each side as per IRC: 3-1983). Table 5 presents the average width of each vehicle category (Society of Indian Automobile Manufacturers, 2025). The mathematical expressions for the weighted average vehicle width and number of virtual lanes are as follows:

$$\text{Weighted average vehicle width} = \frac{\sum N_{cat} \times W_{cat}}{N_{total}}, \quad (4)$$

$$\text{Number of virtual lanes} = \frac{\text{Approach width}}{\text{Weighted average vehicle width} + \text{clearance on both sides}}, \quad (5)$$

Where, N_{cat} is the number of vehicles of each category crossing the stop line simultaneously, W_{cat} is the average vehicle width of each category, and N_{total} is the total number of vehicles passing stop line simultaneously.

To accommodate variations in traffic composition, the number of virtual lanes (parallel movements) is computed at three critical instances: (i) the onset of queue dissipation, (ii) midway through the queue, and (iii) close to the end of queue dissipation. The average of these three values is to be used to represent the number of virtual lanes.

Table 5. Width of different vehicle types

Vehicle type	Width of vehicles (m)
Passenger car, Pickup van	2
Two-wheeler, Motorcycle or Scooter (2W)	0.65
Three-wheeler (Auto-rickshaws (3W)	1.3
Light Commercial Vehicles (LCV)	1.7
Heavy Motor Vehicles (HMV)	2.5

In the absence of field data, a weighted average vehicle width of 1 m can be assumed for two-wheeler-dominated traffic and 2 m for car-dominant traffic. For a 7m approach, this corresponds to approximately four virtual lanes in a two-wheeler dominant scenario and three in a car-dominant scenario.

Calculation of the critical volume and intersection flow ratio

During any green signal phase, a virtual lane or group of virtual lanes on one or more approaches is permitted to move. One of these will have the heaviest traffic, requiring more time than the other lanes active at the same time. If sufficient time is allocated for this lane/lane group, then all other lanes/lane groups will also be well accommodated. Such lane groups are known as critical lane groups, and the sum of volumes in these lanes/lane groups is known as the critical lane volume. In the case of an overlapping phase, the critical path needs to be identified to find out the critical volume. This is done by finding the path through the phase plan that results in the highest possible sum of critical lane volumes. The calculation of the intersection flow ratio is shown in Appendix III. Following the calculation of the optimum cycle length, the effective green time for each approach must be estimated, along with an analysis of the minimum green time for pedestrian movement as a part of comprehensive signal timing design. The procedures for these are detailed in the next section.

3. GREEN PROPORTIONING

The effective green for each phase can be distributed in proportion to the y values as:

$$g = (C_0 - L) \frac{y}{Y} \quad (6)$$

where g – effective green for a phase, sec

L - lost time per cycle, sec

y - critical flow ratio of the phase

Y – intersection flow ratio (sum of all critical flow ratios)

Then, the actual green timing to be provided on the signal head for that phase is:

$$G = g + \text{lost time per phase} - \text{change and clearance interval} \quad (7)$$

4. MINIMUM GREEN BASED ON PEDESTRIAN REQUIREMENT

Pedestrian movement at an intersection can be accommodated either as a concurrent pedestrian phase, where pedestrians cross the street parallel to vehicular movement, or through an exclusive pedestrian phase. They can also be accommodated through lead pedestrian interval (LPI) where pedestrians get a few seconds head start before vehicles move.

The minimum green time required for pedestrian crossing can be calculated using the following equation (HCM, 2000):

$$G_p = 3.2 + \frac{D}{S_p} + 0.81 \frac{N_{ped}}{W_E} \text{ for } W_E > 3m, \quad (8)$$

$$G_p = 3.2 + \frac{D}{S_p} + 0.27N_{ped} \text{ for } W_E \leq 3m, \quad (9)$$

where D - crossing distance (m)

S_p - walking speed of pedestrians (1.2 m/s based on IRC: 93-1985)

N_{ped} - number of pedestrians crossing during an interval

W_E - effective crosswalk width (m)

In the case of concurrent pedestrian phase, the non-conflicting vehicular phase (G+Y) should be more than the above calculated pedestrian green. If the difference is small, green times can be adjusted. Otherwise, a separate pedestrian phase may be considered when pedestrians from all approaches get the right of way simultaneously.

5. PRACTICAL CONSIDERATIONS

- The cycle length should be a multiple of five. If it does not work out to be a multiple of five, it should be rounded up to the next higher multiple of five.
- The maximum cycle length recommended is preferably 120 seconds, which is the recommended maximum acceptable delay for drivers of vehicles and pedestrians.
- A minimum green time for a phase should be 5 seconds.
- It is preferable to limit the number of phases in a cycle to a maximum of four.
- There may be at least three Time of the Day plans prepared for an intersection for weekdays, namely, for (1) morning peak period, (2) afternoon off-peak period, (3) evening peak period. Additionally, weekend and special event plans may be developed.

6. SUMMARISED PROCEDURE FOR SIGNAL DESIGN

1.1 Phase Plan selection

- The selection of a protected or permitted phase plan is first determined based on the threshold values provided in Table 1.
- If a protected phase is required, the next step is to decide between concurrent and lead-lag phasing based on geometric requirements, with concurrent phasing allowed when the diagonal distance between stop lines of opposing approach exceeds 25 m.
- Protected phasing should be further evaluated for the possibility of overlap phasing, which can be adopted if the flow difference between movements within a phase exceeds 60 PCU/h/virtual lane.

- Evaluation of overlap phasing requires estimating the number of virtual lanes, which is calculated by dividing the approach width by the sum of the weighted average vehicle width and side clearances, as given in Equations 4 and 5.
- In the absence of traffic data, a weighted average vehicle width of 1 m for two-wheeler dominant and 2 m for car-dominant traffic can be assumed.

1.2 Determination of yellow and all red time

- After the phase plan is finalised, the yellow and all-red times can be determined using Equation 1.

1.3 Optimum cycle length calculation

- The optimum cycle length is then calculated based on the identified queuing system (M/D/n or M/M/n) and the estimated saturation flow.
- Two-wheeler-dominant scenarios typically follow an M/M/n queuing system, whereas car dominant scenarios follow M/D/n.
- Saturation flow is estimated either by counting the number of vehicles crossing the stop line during queue clearance or, if such data is unavailable, using the empirical value of 900 TCU/h per unit width of a virtual lane.
- Based on the queuing system, the appropriate cycle length formula is selected with values of L obtained from Table 2.
- The critical lane or lane group is then identified to calculate the flow ratio, and the sum of critical flow ratios across phases provides the Y value.

1.4 Green proportioning

- Equations 6 and 7 are used to estimate the effective green time and the actual green time, respectively.

1.5 Pedestrian requirements

- Minimum green can be calculated based on pedestrian requirement as given in Equations 8 or 9.
- Finally, the minimum green time required for pedestrian crossing is calculated using Equations 8 or 9, and $(G+Y)$ of non-conflicting vehicular phase should be higher than this minimum value.

A web-based application for signal design based on these steps is developed and is available at <https://coeut.iitm.ac.in/>. More details about this tool are provided in Appendix IV.

A sample signal timing design is given in Appendix III.

Appendix I

Need for new cycle time expression for Indian traffic condition

Different methods are available for cycle length estimation, and the most prominent among them are Webster's method (Webster, 1958), and Akcelik model (Akcelik, 1981). Webster's method has been extensively used in various countries, including the UK (Transport Research Laboratory, 2025), Canada (Canadian Capacity Guide for Signalised Intersection, 2008), Indonesia (Indonesian Highway Capacity Manual, 1993) and India (IRC: 93-1958). Akcelik method has been extensively used in Australia.

There are several assumptions made in the derivation of these models that won't be satisfied under the heterogeneous and lane-less traffic conditions. For example, the Webster's method assumes uniform vehicle arrivals, constant saturation flow, no queue overflow, fixed lost time and deterministic departures. Also, Webster's equation for cycle time is derived based on a delay equation (Webster, 1958) that has the limitation of estimated delay tending to infinity when the arrival flow approaches capacity, which is not true in practical situations. In other words, Webster's formula overestimates the optimum cycle length as saturation approaches.

While Ackelick's model takes into account overflow delay, it also yields an infinite cycle length when the intersection flow ratio (Y) exceeds one. Moreover, both the models are calibrated under homogeneous and lane-disciplined traffic conditions, which may lead to inaccurate estimates when applied to heterogeneous and lane-less traffic environments. As a result, it is necessary to derive a cycle time formula under non-lane-based mixed traffic conditions relaxing the above assumptions, as discussed next.

Cycle length equations can be derived for various objectives such as delay minimisation, throughput maximisation, queue length minimisation, emission or fuel consumption minimisation, etc. Out of these, delay minimisation is the most common one and is considered in this derivation too. Optimal cycle time derivation for delay minimisation starts with the derivation of a suitable theoretical delay model. In this case, a theoretical delay model explicitly tailored for the heterogeneous, lane-less traffic condition is derived first. The developed delay model effectively captures the characteristic features of HLL traffic, such as the lack of lane discipline, violation of the First-In-First-Out (FIFO) rule, and a substantial mix of vehicle types.

The lack of lane discipline is addressed using the concept of multiple queue-servers in the same lane using the virtual lane concept and the Service in Random Order (SIRO) queue discipline with vacation. SIRO is chosen because, under HLL conditions, smaller vehicles overtake other vehicles in the same lane, and as a result, the First-in-first-out (FIFO) queue discipline is violated. The concept of service with vacation is considered to realistically capture the service happening only during the green interval.

Most of the theoretical delay models have been developed under the assumption of random arrivals and deterministic departures, utilising the M/D/1 queuing model. However, in HLL traffic conditions, lane-free movements lead to multiple parallel vehicle interactions, making a multi-server queueing system more appropriate. Consequently, M/D/n queueing theory is considered.

In addition, analysis of traffic data collected from multiple cities showed that even the departure processes can exhibit randomness under HLL traffic conditions, justifying the applicability of M/M/n queueing theory also for delay estimation. This can be checked by analysing the

departure pattern. The departure distribution can be assessed by counting the number of vehicles discharging during queue clearance, followed by a Chi-square test. In case such data is not available, the following guidelines can be used:

- Under Indian traffic conditions, characterised by HLL traffic and the presence of two-wheelers, the departure distribution is predominantly random (M/M/n).
- In regions with where cars constitute the majority of the traffic composition, the departure distribution tends to be deterministic (M/D/n).

Appendix II

Phase plan selection – Sample calculation

At a four-legged intersection with one through lane and one right-turn lane on each approach, the traffic composition was observed to be 0.45, 0.30, 0.20, and 0.05 for two-wheelers, cars, three-wheelers, and heavy vehicles, respectively. The following right-turning and corresponding opposing through volumes are recorded during the peak hour. Each lane was found to become two virtual lanes. Determine if the right-turn movements on any of the approaches need protection. The stop line-to-stop line distance between opposing approaches was 19m. Suggest a suitable phase configuration for the given data.

Table 6. The RT and opposing TH volumes of the approach

Approach Name	RT volume (veh/h)	Opposing TH volume (veh/h)
North Bound (NB)	600	800
East Bound (EB)	300	1500
South Bound (SB)	550	700
West Bound (WB)	200	2000

Solution:

Step 1: Check for protected vs. permitted phasing

Number of opposing lanes = 1

2/w proportion = 0.45 (in the 40-60% band) – with minimum RT criteria is 575 veh/h (from Table 1)

Check for WB approach

Criterion 1: RT volume = 200 < 575 (Table value) – no need to protect

Criterion 2: Opposite through volume = 2000; corresponding minimum RT x TH = 575000 (From Table 1)

Actual RT x TH = 200 x 2000 = 400000 < 575000 – no need to protect

Check for EB approach

Criterion 1: RT volume = 300 < 575 (Table value) – no need to protect

Criterion 2: Opposite through volume = 1500; corresponding minimum RT x TH = 575000

Actual RT x TH = 300 x 1500 = 450000 < 575000 – no need to protect

Check for SB approach

Criterion 1: RT volume = 550 < 575 (Table value) – no need to protect

Criterion 2: Opposite through volume = 700; corresponding minimum RT x TH = 575000

Actual RT x TH = 550 x 700 = 385000 < 575000 – no need to protect

Check for NB approach

Criterion 1: RT volume = 600 > 575 (Table value) – need to protect

Since north bound right turn is protected, the south bound also gets protected. However, the right turn movements are permitted in the EB and WB approach, resulting in a three-phase configuration.

Step 2: Check for split vs. concurrent and lead-lag phasing

Since the NB and SB right-turn movements are protected, it is necessary to evaluate the suitability of split, concurrent, and lead-lag phasing options, as discussed below.

The distance between stop lines in both N-S and E-W directions is 19 m.

To examine the feasibility of concurrent phasing in the protected phasing of the NB and SB approach,

The diagonal distance connecting the turning centers of NB – SB approaches =

$$\sqrt{19^2 + 19^2} = 26.9 \text{ m}$$

Since this distance is more than the recommended minimum of 25 m, either concurrent or lead-lag phasing is feasible for the RT movements of N-S approach. Given the presence of a separate right-turn lane, concurrent phasing is the appropriate choice. Within concurrent phasing, the selection between leading or lagging phases depends on the relative through and right-turn volumes. As the through volumes exceed the right-turn volumes, a lagging right-turn phase is recommended.

Step 3: Check for overlap vs. non-overlap phasing

The assessment of overlap phasing is necessary only for the NB and SB movements, as the EB and WB turns are not protected. To evaluate the possibility of implementing overlap phasing, the traffic volumes in PCU/h per virtual lane for the NB and SB approaches were compared. For this purpose, the RT and TH volumes for the NB approach were converted to PCU/h, considering two virtual lanes for each movement, and the results are presented in Table 7.

Table 7. The RT and TH volumes of the NB and SB approach

Approach	RT Volume*						TH volume					
	Volume (veh/h)	2W	Car	3W	HMV	Volume** (PCU/h/virtual lane)	Volume (veh/h)	2W	Car	3W	HMV	Volume (PCU/h/virtual lane)
NB	600	270	180	120	30	296	700	315	210	140	35	345.6
SB	550	248	165	110	27	272	800	360	240	160	40	395

*Based on given proportions

**Multiplying with PCU and dividing with number of virtual lanes

From the table, it can be observed that the RT volumes for the NB and SB approaches are 296 and 272 PCU/h per virtual lane, respectively, while the TH volumes for the NB and SB approaches are 345.6 and 395 PCU/h per virtual lane. Since the difference between the volumes in the south and northbound approaches does not exceed 60 PCU/h per virtual lane, it indicates that overlap phasing is not required.

The resulting phase plan for the given traffic condition is presented in Figure 9.

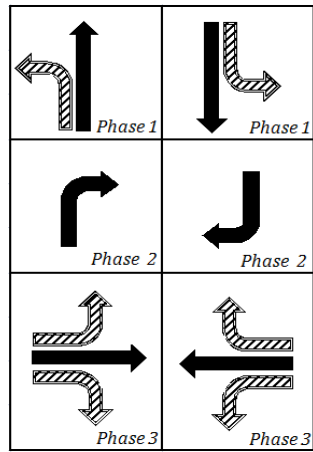


Figure 9 Phase plan for the prevailing traffic condition

Appendix III

Signal Timing Design – Sample calculation

Data: A four-legged intersection with a six-lane major road and a four-lane minor road crossing at level grade. Peak hour volumes (in PCU/hour) on different approaches are shown in Figure 10. In the field, traffic composition consists of 86% two-wheelers, 9% cars, 4% three-wheelers, and the remaining percentage heavy vehicles. The data on arrival and departure revealed that both follow a random distribution.

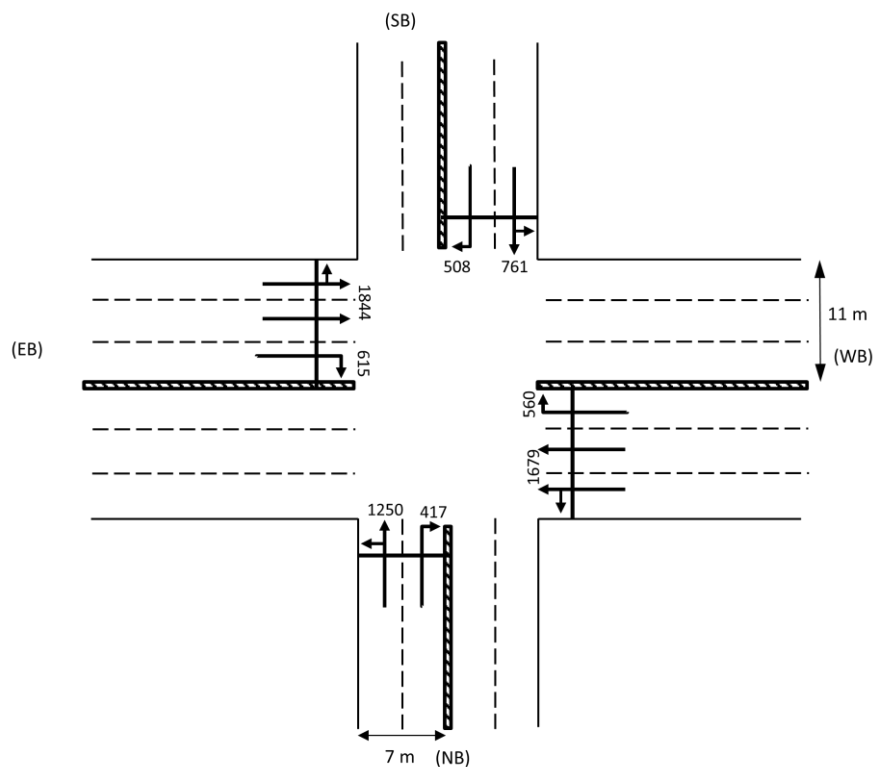


Figure 10 Traffic and geometric details of the field site

Additional details for the analysis are as follows:

- Reaction time of drivers - 0.5 s.
- Average approach speed of the vehicle – 30 km/h
- Acceleration of the vehicle – 3 m/s^2
- Average length of standard vehicle - 6m
- Side clearance of both sides of the vehicle - 0.53 m.
- The pedestrian volumes during peak hour are 80 and 130 for N-S and E-W directions, respectively
- Pedestrian crosswalk width - 2.5 m
- The walking speed of pedestrians is 1.2 m/s.
- Distance between stop lines in the N-S direction is 24 m and E-W direction is 17 m.

The number of vehicles crossing the stop line at three instances of queue clearance for one approach is given in Table 8. Use the vehicle width provided in Table 5 for the virtual lane calculation.

Table 8. The classified vehicle counts crossing the stop line of the WB approach at three instances of queue clearance

Time instances	Number of vehicles concurrently crossing the stop line				
	2W	3W	Car	LCV	HMV
Beginning of the queue (during the first 30% of green)	5	0	1	1	0
Middle of the queue (during the second 30% of green)	4	1	0	0	1
End of the queue	4	0	1	0	0

The classified count of vehicles crossing the stop line during the initial period of queue dissipation (5-second interval) after the initial lost time is provided in Table 9.

Table 9. The classified vehicle counts crossing the stop line during queue dissipation of WB approach

Time for queue clearance (s)	Through Movement					Right turn				
	2W	3W	car	LCV	HMV	2W	3W	car	LCV	HMV
25	54	3	9	2	1	16	0	2	0	0

Step 1: Phase plan Design

Step 1a: Check for protected vs. permitted phasing

To determine the phase plan, the through and right-turning volumes are required, and these volumes are summarised in Table 10.

Table 10. The turning volumes of each approach

The through and right turn volume (veh/h)							
NB		SB		EB		WB	
TH	RT	TH	RT	TH	RT	TH	RT
1250	417	761	508	1844	615	1679	560

Since the EB and WB approaches have more than two lanes, a protected phasing plan is required for these directions. The NB and SB approaches have two lanes and contain approximately 86% two-wheelers. Therefore, the need for protected phasing is evaluated using Table 1, and the results are presented in Table 11.

Table 11. Check for protected vs. permitted phasing in the NB and SB approach

Criteria	NB approach			SB approach		
	Field	Threshold	Remarks	Field	Threshold	Remarks
Right turn volume	417	400	RT exceeds the threshold value. Hence, protection is required	508	400	Both RT and cross product exceeds the threshold value. Hence, needs protection.
Cross product of right turn and opposing through volume	=417x761=317337	5,50,000 (From Table 1, for the case with > 85% bikes and two opposing through lanes)		=508x1250=635000	5,50,000 (From Table 1, for the case with > 85% bikes and two opposing through lanes)	

Owing to the need for protected movements on all approaches, a four-phase signal configuration is adopted.

Step 1b: Check for concurrent vs. lead-lag phasing

For the EW and NB directions, the distance between stop lines is 17 and 24m, respectively. Then, the diagonal distance connecting the turning centres of opposing right turn movement is

$$\sqrt{24^2 + 17^2} = 29.4 \text{ m.}$$

Since this distance is more than the recommended minimum of 25 m, either concurrent or lead-lag phasing is feasible. Given the presence of a separate right-turn lane, concurrent phasing is the appropriate choice. Within concurrent phasing, the selection between leading or lagging phases depends on the relative through and right-turn volumes. As the through volumes exceed the right-turn volumes, a lagging right-turn phasing is recommended for N-S and E-W right turn movements.

Step 1c: Check for overlap vs. non-overlap phasing

This analysis first requires the estimation of the number of virtual lanes for each approach, which is calculated as described below.

Virtual lane calculation

Using the data given in Table 9, the number of virtual lanes is calculated for the WB approach, and the results are presented in Table 12.

Table 12. Calculation of the number of virtual lanes for the WB approach

Time instances	Number of vehicles crossing the stop line					Weighted average vehicle width*	Number of virtual lanes**	Average number of virtual lanes
	2W	3W	Car	LCV	HMV			
Beginning of the queue	5	0	1	1	0	0.99	5.4	5.4~6
Middle of the queue	4	1	0	0	1	1.07	5.2	
End of the queue	4	0	1	0	0	0.92	5.6	

The sample calculation of the weighted average vehicle width at the beginning of the queue is:

$$\text{*Weighted average vehicle width} = \frac{\sum \text{number of vehicles of each category} \times \text{vehicle width}}{\text{Total number of vehicle passing stop bar}} = \frac{(5 \times 0.65 + 1 \times 2 + 1 \times 1.7)}{7} = 0.99$$

$$\text{**Number of virtual lanes} = \frac{\text{Total approach width}}{\text{Weighted average vehicle width} + \text{clearance on both sides}} = \frac{11}{0.99 + 1.06} = 5.4$$

In the WB approach, there were three lanes in the field, which effectively functioned as six virtual lanes, as shown in the table above. Thus, each lane has become two virtual lanes. Since data on other approaches was not available, and the vehicle composition is the same across all approaches, the same factor is applied in other approaches. Based on this, the number of virtual lanes for SB, EB, and NB is taken as 4, 6, and 4, respectively.

Check for overlap phasing

To assess the possibility of overlap phasing, the volumes in PCU/h per virtual lane were estimated, based on the PCU values from IRC: 106-1990. The values obtained for the EB and WB approaches, and similarly for the NB and SB approaches, are then compared to identify potential overlaps. Virtual lane analysis indicates six virtual lanes in both EB and WB directions, and four lanes in NB and SB

directions. Among these, right turning movements are considered from the two rightmost virtual lanes, while through movements are assigned from the remaining lanes. The results are presented in Table 13.

Table 13. The RT and TH volumes of the EB and WB approach

Approach name	RT Volume*						TH volume					
	Volume (veh/h)	2W	Car	3W	HMV	Volume (PCU/h/virtual lane) **	Volume (veh/h)	2W	Car	3W	HMV	Volume (PCU/h/virtual lane)
NB	417	358	38	17	4	168	1250	1075	113	50	13	503
SB	508	437	46	20	5	204	761	655	69	30	8	306
EB	615	528	55	25	6	247	1844	1585	166	74	18	371
WB	560	481	50	22	6	225	1679	1444	151	67	17	338

*Based on given proportions

**Multiplying by PCU and dividing by the number of virtual lanes

Since it is a concurrent phasing, the RT of the opposing approach and the TH movements of the opposing approach will happen simultaneously, and they should be compared for the possibility of overlap checking. From the table, it can be observed that the RT volumes for the NB and SB approaches are 168 and 204, respectively, and similarly, TH volumes are 503 and 306, respectively. Since the difference in TH volume exceeds 60 PCU/h/virtual lane, overlap phasing is recommended. Similarly, the RT volumes for the EB and WB approaches are 247 and 225 PCU/h/virtual lane, respectively, while the TH volumes for the EB and WB approaches are 371 and 338 PCU/h/virtual lane, respectively. Since the difference in volume of the movement that occur in a phase does not exceeds 60 PCU/h/virtual lane in both cases, overlap phasing is not recommended. The corresponding phase plan is presented in Figure 11.

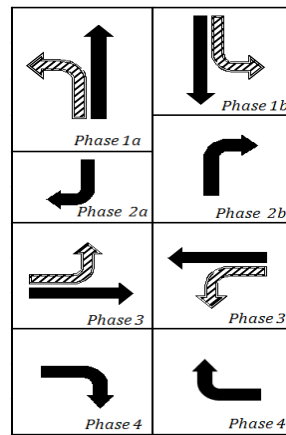


Figure 11 The phase plan for the prevailing traffic condition

Step 2: Calculation of change and clearance interval

Using Equation 1, the minimum change and clearance time is:

$$T_{WB \& EB} = 0.5 + \frac{30 \cdot (5/18)}{2(3+9.8 \cdot 0.0)} + \frac{22+6}{30 \cdot (5/18)} = 5.24 \text{ seconds}$$

$$T_{NB \& SB} = 0.5 + \frac{30 \cdot (5/18)}{2(3+9.8 \cdot 0.0)} + \frac{14+6}{30 \cdot (5/18)} = 4.28 \text{ seconds}$$

Based on this, a change and clearance interval of 5 seconds is adopted for WB and EB, and 4 seconds for NB and SB.

Step 3: Calculation of optimum cycle length

Determination of the optimum cycle length requires identification of the appropriate queueing system. Since the traffic stream is predominantly composed of two-wheelers, an M/M/n queueing model is adopted, and Equation (2) is used to determine the optimum cycle length. This calculation requires estimating saturation flow, determining the number of virtual lanes, and computing the intersection flow ratio. The estimation of virtual lanes is described in Step 1, while the procedures for estimating the remaining variables are discussed below.

Step 3a: Calculation of saturation flow

Using the data given in Table 9, the saturation flow is calculated and presented in Table 14. The vehicular counts are converted into PCU values using the equivalency factors specified in IRC 106:1990. Then, the turning volumes are converted to through equivalent volumes using Table 4. The total TCU for the queue clearance time is calculated, and these values are then scaled to an hourly rate. This hourly rate is subsequently divided by the number of virtual lanes to obtain the saturation flow, expressed in PCU per hour of green time per virtual lane, which came to be 1750 PCU per hour of green per virtual lane. Since the composition is the same across all the approaches, this value is also applied to the other approaches.

The sample calculation of total TCU for the first 5-second interval is:

$$\text{Total TCU} = (54 \times 0.75 + 3 \times 1.2 + 9 \times 1 + 2 \times 1.4 + 1 \times 2.2) + 1.05 \times (16 \times 0.75 + 2 \times 1) = 72.80$$

Table 14. Saturation flow calculation of the WB approach

Time for queue clearance (s)	Through Movement					Right turn					Total TCU	Avg saturation flow of the approach (TCU/h)	Saturation flow (TCU/h/virtual lane)
	2W	3W	car	LCV	HMV	2W	3W	car	LCV	HMV			
25	54	3	9	2	1	16	0	2	0	0	72.80	10483.2	1747.2 ≈ 1750

Step 3c: Estimation of intersection flow ratio and optimum cycle length

To estimate the intersection flow ratio(Y), the first step is to identify the virtual lane group. From the schematic, turning vehicles are sharing the lanes with through vehicles. Hence, all lanes in the approach are considered as a single-lane group. The flow ratios (y) are then calculated for this virtual lane group and used in the estimation of Y. The detailed computation of Y is presented in Table 15.

Table 15. Calculation of intersection flow ratio

Phases	Phase 1&2				Phase 3		Phase 4	
Approach	NB		SB		EB	WB	EB	WB
Movements	TH	RT	TH	RT	RT	RT	TH	TH
Flow per virtual lane (PCU/h/virtual lane)	503	168	306	204	247	225	371	338
Saturation flow per virtual lane (TCU/green/virtual lane)	1750	1750	1750	1750	1750	1750	1750	1750
y= flow/saturation flow	0.287	0.096	0.175	0.117	0.141	0.129	0.212	0.193

Critical y of each phase (path based as it is an overlap phase)	Higher of (0.287+0.117) and (0.175+0.096) = 0.404	Higher of 0.141 and 0.129 = 0.141	Higher of 0.212 and 0.193=0.212
$Y=\sum y$	0.758		

Given that, both arrivals and departures exhibit a random distribution pattern, the cycle length should be determined using the expression provided for an M/M/n queueing system. The corresponding cycle length equation is:

$$C_0 = \frac{1.9 L + 3.2}{1 - Y}$$

Given the predominant presence of two-wheelers in the traffic composition, protected phasing, and peak traffic conditions, the lost time per phase is considered to be 2.16 seconds (Table 2). For a four-phase signal design, this results in a total lost time of 8.64 seconds per cycle. Hence, the optimum cycle length is:

$$C_0 = \frac{1.9 L + 3.2}{1 - Y} = \frac{1.9 \times 8.64 + 3.2}{1 - 0.76} = 85 \text{ seconds}$$

Step 4: Green proportioning of each approach

The effective green and actual green timings are calculated using Equations 5 and 6, respectively, and values are presented in Table 16. The corresponding timing diagram is shown in Figure 11.

Table 16. Calculation of cycle length and green proportioning

Phases	Phases 1 and 2				Phase 3		Phase 4	
Approach	NB		SB		EB	WB	EB	WB
Movements	TH	RT	TH	RT	RT	RT	TH	TH
Cycle length	85							
Effective green of each phase $g = (C - L) \frac{y}{Y}$	$g_{1 \text{ and } 2} = (85 - 8.64) \frac{0.404}{0.758} = 41$				$g_3 = (85 - 8.64) \frac{0.141}{0.758} = 14$		$g_4 = (85 - 8.64) \frac{0.212}{0.758} = 21$	
Effective green of each movement	29	14	27	12	14		21	
Actual green timing G=g+ lost time per phase- change interval	27	13	25	10	11		18	

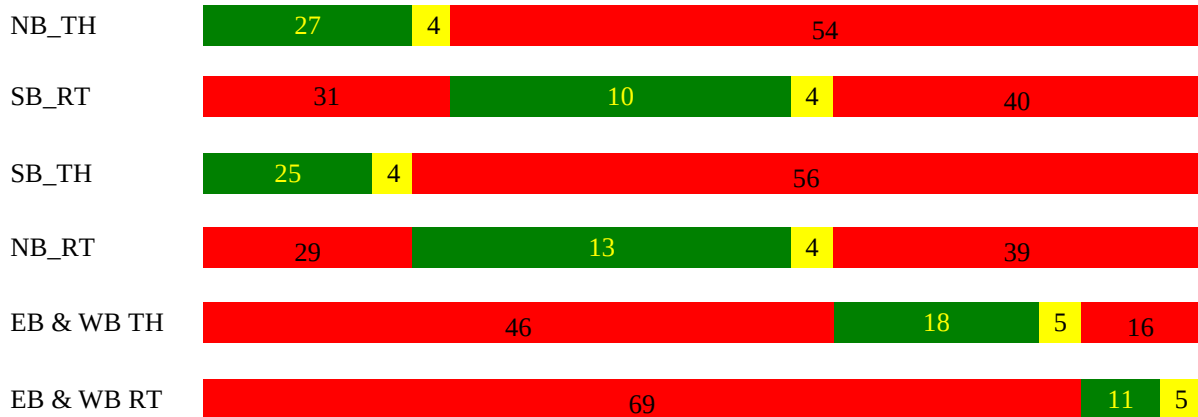


Figure 11 The timing diagram of signal design

Step 6: Check for pedestrian requirements

The minimum green based on the pedestrian requirement is calculated using Equation 9 and the results are presented in Table 17. The final timing plan in the ring and barrier diagram is illustrated in Figure 18.

Table 17. Check for pedestrian requirements

Direction of pedestrian movement	Minimum pedestrian green time required	Green time available for safe crossing of the for pedestrians	Inference
N-S direction	$3.2 + (11 \times 2/1.2) + (0.27 \times (80 \times 85/3600)) = 22 \text{ s}$	For EB approach, $25+4=29\text{s}$ For WB approach, $27+4=31\text{s}$	The available green time for safe crossing of pedestrians is higher than the minimum green required for pedestrian crossing. Hence, no need for a separate pedestrian phase
E-W direction	$3.2 + (7 \times 2/1.2) + (0.27 \times (130 \times 85/3600)) = 16 \text{ s}$	NB and SB approach, $18+5= 23\text{s}$	

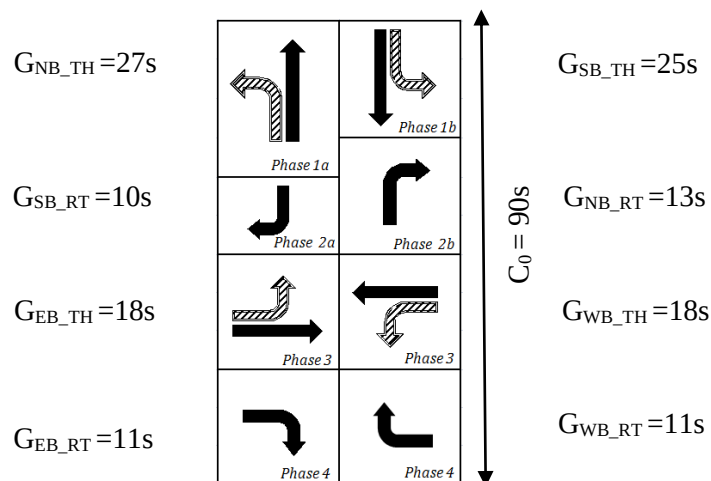


Figure 12 Ring diagram with timings of each movement

Appendix IV

Cyclo-Help: A web application for signal design

User Guide: Step-by-Step

CYCLE Length Optimisation for HETerogeneous Lane-less Paradigm (Cyclo-Help) is a tool designed for traffic engineers and planners, students and researchers, and consultants who are interested in signal design. It offers a robust platform for designing signal timings at isolated intersections, specifically calibrated for heterogeneous and less-lane-less (HLL) traffic conditions. Such conditions are typical in regions where vehicle fleets are highly diverse (with significant proportions of two- and three-wheelers) and lane discipline is not strictly maintained. This will work for the traditional lane-based and homogeneous traffic conditions as well.

The Cyclo-Help can be accessed through the following link: <https://coeut.iitm.ac.in/>. The user interface opens with a Welcome Page as shown in Figure 13, with two options to choose from, i) to directly start the design of a fixed-time signal, the details of which are discussed next, and ii) the documentation page that provides an overview of the tool, including detailed instructions on how to use the application and an explanation of the signal design.



Figure 13 User interface of Cyclo-Help

Design of fixed-time signal

Step 1: Select the location of the intersection

You can select the location in two ways:

1) Using Latitude and Longitude

- Navigate to the desired intersection on a map.
- Right click on the intersection, copy the latitude and longitude, and paste them into the box provided on the web page.

2) Using the OpenStreetMap Option

- Select the intersection directly from the OpenStreetMap provided on the web page.
- Once you click on the location, the latitude and longitude will be automatically selected. The signal design will then be generated for that location.

After selecting the location, click Next to proceed to Field Calibration.

Step 2: Configure Intersection Parameters (Field Calibration)

The following details are required for designing the signal timings, and Figure 14 provides a visual guide to assist in entering the corresponding input data into the software.

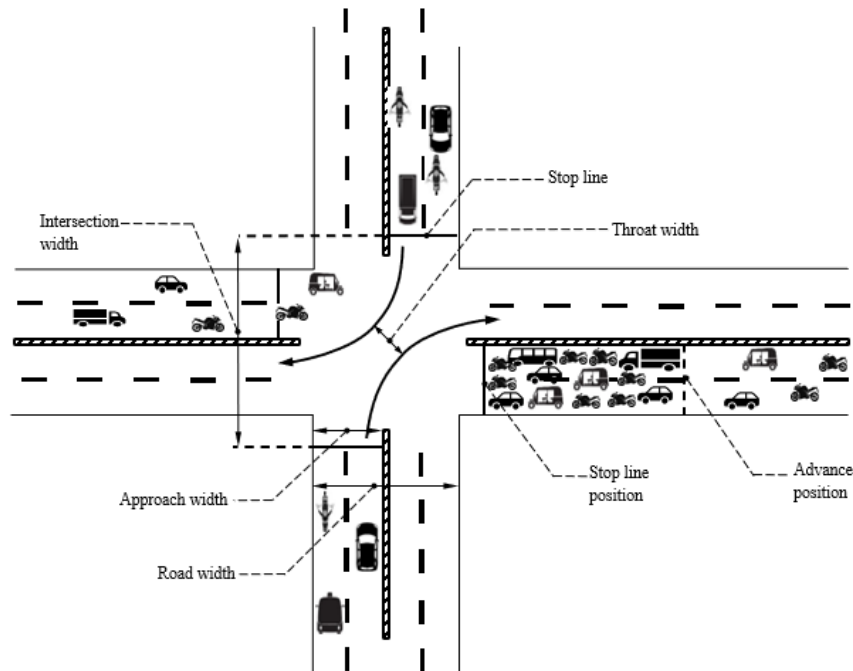


Figure 14 Illustration of intersection parameters

- 1) **Type of intersection**
 - Choose whether the intersection is 3-legged or 4-legged.
- 2) **Select the missing approach**
 - By default, the northbound approach is set as missing.
- 3) **Exclusive right-turn and left-turn lanes**
 - Indicate whether exclusive right-turn and/or left-turn lanes are provided
 - By default, this option is set to No.
- 4) **Width of Approach (in meters)**
 - Enter the width for each approach.
 - By default, all approaches are set to 7 meters.
 - Input field corresponding to the missing approach is disabled.
- 5) **Distance from Stop Line to Stop Line (in meters)**
 - Enter the distance from stop line to stop line along the North - South direction and the East – West direction.
 - If this data is not provided, the application automatically estimates the value based on the specified approach widths.
- 6) **Class-wise Peak 15-Minute Count (At advance position)**
 - Enter the number of vehicles in each class for the peak 15 minutes, for each approach.
 - For 3-legged intersections, the missing approach input will be disabled.
- 7) **Class-wise Peak 15-Minute Right-Turn Count (At Stop line position)**

- Enter the right-turn counts by vehicle class for the peak 15 minutes, for each approach.
 - For 3-legged intersections, counts will be disabled for the missing approach and for right turns into that approach.
- 8) **Class-wise Peak 15-Minute Through and Left-Turn Count (At Stop Line position)**
- To improve accuracy, enter the through and left-turn counts by vehicle class for the peak 15 minutes.
 - Based on these counts and the right-turn counts, the software will calculate the turning proportions and apply them to the counts at the advance position.
 - If these values are not entered, the through and left-turn counts will be derived by subtracting the right-turn count from the advance position.
 - For 3-legged intersections, the missing approach input will be disabled.
- 9) **Pedestrian Crossing Count (Peak 15 Minutes)**
- Enter the number of pedestrians crossing during the peak 15 minutes.
 - Input will be disabled for the missing approach in a 3-legged intersection.
- 10) **Average Operating Speed (km/h)**
- Enter the average operating speed for each approach.
 - If unavailable, enter the posted speed limit.
 - Input will be disabled for the missing approach in a 3-legged intersection.

After entering the required data, users are advised to click the “Save” button provided at the end for saving the user input values during this session before clicking the “Analyse” button.

Step 3: Run Analysis

Once all required information has been entered, the system will generate:

- An intersection map of the location for which the signal is designed.
- A ring diagram illustrating the phase plan.
- A timing diagram corresponding to the phase plan.

If the minimum requirements for designing a phase plan are not satisfied, then an error message will be displayed instead of the above.

Step 4. Troubleshooting

Error 1: " Intersection flow ratio (Y) ≥ 1 for the input values"

This is the most common error. It means that the traffic demand you have entered is greater than the physical capacity of the intersection ($\sum v/s \geq 1$). The delay in this scenario is theoretically infinite, as the design is only for under saturated ($v/s < 1$) condition.

Solution: You may increase the intersection capacity by increasing approach width or by adding more lanes. Error 2: “Phase 1 green time not meeting the minimum green criteria”

In case the calculated green time is less than the minimum green time based on pedestrian requirements, an error message similar to the one above will be displayed and the signal has to be re-designed. The message displayed will vary based on the phase for which the minimum requirement is not met.

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